$= P(t) \cdot n\, C_{n-1}^{\ldots}$ $\quad (k-1)!(n-k)!\,(F(t))\cdot(1-F(t))\cdot P(t)$

$\boxed{№3.}$

$$P(x) = \begin{cases} \dfrac{e^{-x/\theta}}{\theta}, & x \geq 0 \\ 0, & x < 0 \end{cases} \qquad \theta > 0 \qquad n = 3$$

$\vec{x}_n$ - выборка

$\tilde{\theta}_1 = \bar{x}$

$\tilde{\theta}_3 = x_{(2)}$

$$M\xi = \int_{-\infty}^{+\infty} x\, dF(x,\theta) = \int_0^{+\infty} x \frac{1}{\theta} e^{-\frac{x}{\theta}} dx = -\frac{\theta}{\theta}\int_0^{+\infty} x\, d e^{-\frac{x}{\theta}} =$$

$$= -x e^{-\frac{x}{\theta}}\Big|_0^{+\infty} + \int_0^{+\infty} e^{-\frac{x}{\theta}} dx = -\theta e^{-\frac{x}{\theta}}\Big|_0^{+\infty} = \theta$$

$$M\xi^2 = \int_{-\infty}^{+\infty} x^2\, dF(x,\theta) = \int_0^{+\infty} x^2 \frac{1}{\theta} e^{-\frac{x}{\theta}} dx = -\int_0^{+\infty} x^2 d e^{-\frac{x}{\theta}} =$$

$$= -x^2 e^{-\frac{x}{\theta}}\Big|_0^{+\infty} + \int_0^{+\infty} e^{-\frac{x}{\theta}} \cdot 2x\, dx = -2\theta \int_0^{+\infty} x\, d e^{-\frac{x}{\theta}} = -2\theta x e^{-\frac{x}{\theta}}\Big|_0^{+\infty} +$$

$$+ 2\theta \int_0^{+\infty} e^{-\frac{x}{\theta}} dx = -2\theta^2 e^{-\frac{x}{\theta}}\Big|_0^{+\infty} = 2\theta^2$$

$$D\xi = M\xi^2 - (M\xi)^2 = 2\theta^2 - \theta^2 = \theta^2$$

а) Проверка на несмещённость

① $\tilde{\theta}_1 = \bar{x}$

симметрии. ... функцию распределения, гистограмму и ра θ. Сгенерируй

$$M\tilde{\theta}_1 = M\left(\frac{1}{n}\sum_{k=1}^{n} x_k\right) = \frac{1}{n}\cdot n\, M\xi = M\xi = \theta$$

$\Rightarrow \tilde{\theta}_1$ - несмещённая оценка

② $\tilde{\theta}_3 = x_{(2)}$

$$\xi_{(k)} \sim \sum_{i=k}^{n} C_n^i \left(F(t)\right)^i \left(1-F(t)\right)^{n-i}$$

$$P_{x_{(k)}}(t) = \frac{n!}{(k-1)!\,(n-k)!}\left(F(t)\right)^{k-1}\left(1-F(t)\right)^{n-k}\cdot P(t) = \langle n=3,\ k=2\rangle =$$

$$= \frac{3!}{1!\,1!}\left(F(t)\right)\left(1-F(t)\right)\cdot P(t)$$

$$F(t) = \int_{-\infty}^{t} P(x)\,dx = \int_0^t \frac{1}{\theta} e^{-\frac{x}{\theta}}\,dx = -e^{-\frac{x}{\theta}}\Big|_0^t = -e^{-\frac{t}{\theta}} + 1$$

$$M\tilde{\theta}_3 = \int_{-\infty}^{+\infty} x\cdot P_{x_{(k)}}(x)\,dx = \int_0^{+\infty} 6x\left(1-e^{-\frac{x}{\theta}}\right)e^{-\frac{x}{\theta}}\cdot\frac{1}{\theta}e^{-\frac{x}{\theta}}\,dx =$$

$$= \frac{6}{\theta}\int_0^{+\infty} x e^{-\frac{2}{\theta}x}\,dx - \int_0^{+\infty}\frac{6}{\theta}x e^{-\frac{3}{\theta}x}\,dx = -3\int_0^{+\infty} x\,d e^{-\frac{2}{\theta}x} +$$

$$+ 2\int_0^{+\infty} x\,d e^{-\frac{3}{\theta}x} = -3xe^{-\frac{2}{\theta}x}\Big|_0^{+\infty} + 3\int_0^{+\infty} e^{-\frac{2}{\theta}x}\,dx + 2xe^{-\frac{3}{\theta}x}\Big|_0^{+\infty} -$$

$$-2\int_0^{+\infty} e^{-\frac{3}{\theta}x}\,dx = -\frac{3\theta}{2}e^{-\frac{2}{\theta}x}\Big|_0^{+\infty} + \frac{2\theta}{3}e^{-\frac{3}{\theta}x}\Big|_0^{+\infty} = \frac{3\theta}{2} - \frac{2\theta}{3} =$$

$$= \frac{9-4}{6}\theta = \frac{5\theta}{6}$$

$\Rightarrow \tilde{\theta}_3$ не является несмещённой

$$\tilde{\theta}_3' = \frac{6}{5}x_{(2)}$$

$$M\tilde{\theta}_3' = \frac{6}{5}M x_{(2)} = \frac{6}{5}\cdot\frac{5\theta}{6} = \theta$$

$\Rightarrow \tilde{\theta}_3'$ является несмещённой

в) Какая из исправленных оценок более эффективна?

$\tilde{\theta}_1 = \bar{x}$ $\qquad D\tilde{\theta}_1 = D\left(\frac{1}{n}\sum_{k=1}^{n} x_k\right) = \frac{1}{n^2}\, n\, D\xi = \frac{\theta^2}{n} = \frac{\theta^2}{3}$

$$\tilde{\theta}_3' = \frac{6}{5} x_{(2)}$$

$$M x_{(2)} = \frac{5\theta}{6}$$

$$M(x_{(2)})^2 = \int_{-\infty}^{+\infty} x^2 P_{x_{(2)}}(x)\,dx = \int_0^{+\infty} 6x^2\left(1-e^{-\frac{x}{\theta}}\right)e^{-\frac{x}{\theta}}\cdot\frac{1}{\theta}e^{-\frac{x}{\theta}}dx =$$

$$= \frac{6}{\theta}\int_0^{+\infty} x^2 e^{-\frac{2}{\theta}x}dx - \int_0^{+\infty}\frac{6}{\theta}x^2 e^{-\frac{3}{\theta}x}dx = -3\int_0^{+\infty} x^2 d e^{-\frac{2}{\theta}x} + 2\int_0^{+\infty} x^2\cdot d e^{-\frac{3}{\theta}x} =$$

$$= -3x^2 e^{-\frac{2}{\theta}x}\Big|_0^{+\infty} + 3\int_0^{+\infty} 2x e^{-\frac{2}{\theta}x}dx + 2x^2 e^{-\frac{3}{\theta}x}\Big|_0^{+\infty} - 2\int_0^{+\infty} 2x e^{-\frac{3}{\theta}x}dx =$$

$$= -3\theta x e^{-\frac{2}{\theta}x}\Big|_0^{+\infty} + 3\theta\int_0^{+\infty} e^{-\frac{2}{\theta}x}dx + \frac{4\theta}{3}x e^{-\frac{3}{\theta}x}\Big|_0^{+\infty} - \frac{4\theta}{3}\int_0^{+\infty} e^{-\frac{3}{\theta}x}dx =$$

$$= -\frac{3\theta^2}{2}e^{-\frac{2}{\theta}x}\Big|_0^{+\infty} + \frac{4\theta^2}{9}e^{-\frac{3}{\theta}x}\Big|_0^{+\infty} = \frac{3\theta^2}{2} - \frac{4\theta^2}{9} = \frac{19\theta^2}{18}$$

$$D x_{(2)} = M(x_{(2)})^2 - (M x_{(2)})^2 = \frac{19\theta^2}{18} - \frac{25\theta^2}{36} = \frac{13\theta^2}{36}$$

$$D\tilde{\theta}_3' = D\left(\frac{6}{5}x_{(2)}\right) = \frac{36}{25}\cdot D x_{(2)} = \frac{36}{25}\cdot\frac{13\theta^2}{36} = \frac{13}{25}\theta^2$$

$$D\tilde{\theta}_1 < D\tilde{\theta}_3'$$

$$\frac{\theta^2}{3} < \theta^2\cdot\frac{13}{25} \quad \Big|\cdot\frac{75}{\theta^2}$$

$$25 < 39$$

$\Rightarrow$ $\tilde{\theta}_1$ более эффективна, чем $\tilde{\theta}_3'$

c) Исследовать на эффективность с помощью неравенства Крамера - Рао

регулярность модели

a) $P(x) = \begin{cases} \frac{1}{\theta}e^{-\frac{x}{\theta}}, & x \geq 0 \\ 0, & x < 0 \end{cases}$ дифф по θ на $(0; +\infty)$

б) $\frac{\partial}{\partial\theta}\underbrace{\int dF}_{=1} = \int d\,\frac{\partial F}{\partial\theta} \Rightarrow \int \frac{\partial p}{\partial\theta}dx = 0$

$$\int_0^{+\infty}\frac{\partial}{\partial\theta}\left(\frac{1}{\theta}e^{-\frac{x}{\theta}}\right)dx = \int_0^{+\infty}\left(-\frac{1}{\theta^2}e^{-\frac{x}{\theta}} + \frac{x}{\theta^3}e^{-\frac{x}{\theta}}\right)dx =$$

$$= \frac{1}{\theta^2}\int_0^{+\infty}\frac{x}{\theta}e^{-\frac{x}{\theta}}dx - \frac{1}{\theta^2}\int_0^{+\infty}e^{-\frac{x}{\theta}}dx = -\frac{1}{\theta^2}\int_0^{+\infty}x\,de^{-\frac{x}{\theta}} - \int_0^{+\infty}\frac{1}{\theta^2}e^{-\frac{x}{\theta}}dx =$$

$$= -\frac{1}{\theta^2}xe^{-\frac{x}{\theta}}\Big|_0^{+\infty} + \frac{1}{\theta^2}\int_0^{+\infty}e^{-\frac{x}{\theta}}dx - \int_0^{+\infty}\frac{1}{\theta^2}e^{-\frac{x}{\theta}}dx = 0$$

в) $I(\theta) = M\left[\left(\frac{\partial \ln p(x,\theta)}{\partial\theta}\right)^2\right] = \int_0^{+\infty}\left(\frac{\partial \ln p(x,\theta)}{\partial\theta}\right)^2 p(x,\theta)dx$

$$\frac{\partial}{\partial\theta}\left(\ln\left(p(x,\theta)\right)\right) = \frac{\partial}{\partial\theta}\left(\ln\frac{1}{\theta} + \left(-\frac{x}{\theta}\right)\right) = \frac{\partial}{\partial\theta}\left(-\ln\theta - \frac{x}{\theta}\right) =$$

$$= -\frac{1}{\theta} + \frac{x}{\theta^2}$$

$$I(\theta) = \int_0^{+\infty}\left(-\frac{1}{\theta} + \frac{x}{\theta^2}\right)^2\frac{1}{\theta}e^{-\frac{x}{\theta}}dx = \int_0^{+\infty}\frac{1}{\theta}\frac{(x-\theta)^2}{\theta^4}e^{-\frac{x}{\theta}}dx =$$

$$= \frac{1}{\theta^5}\int_0^{+\infty}x^2e^{-\frac{x}{\theta}}dx - \frac{2}{\theta^4}\int_0^{+\infty}xe^{-\frac{x}{\theta}}dx + \frac{1}{\theta^2}\int_0^{+\infty}e^{-\frac{x}{\theta}}dx =$$

$$= -\frac{1}{\theta^4}x^2e^{-\frac{x}{\theta}}\Big|_0^{+\infty} + \frac{1}{\theta^4}\int_0^{+\infty}2xe^{-\frac{x}{\theta}}dx - \frac{2}{\theta^4}\int_0^{+\infty}xe^{-\frac{x}{\theta}}dx + \frac{1}{\theta^3}\cdot\left(-\theta e^{-\frac{x}{\theta}}\right)\Big|_0^{+\infty} =$$

$= -\frac{1}{\theta^2}e^{-\frac{x}{\theta}}\Big|_0^{+\infty} = \frac{1}{\theta^2}$ непр на $(0;+\infty)$ и $I(\theta)>0$

$\Rightarrow$ модель регулярна

① $\tilde{\theta}_1$ регулярность оценки

а) модель регулярна +

б) несмещённость оценки +

в) $D\tilde{\theta}_1 = \frac{\theta^2}{3}$ ограничена на $\forall$ компакте из $(0; +\infty)$

$\Rightarrow$ оценка $\tilde{\theta}_1$ является регулярной

Достаточное условие эффективности

$$\left.\begin{array}{l} D\tilde{\theta}_1 = \frac{\theta^2}{3} \\ \frac{1}{n \cdot I(\theta)} = \frac{1}{n} \cdot \theta^2 = \frac{\theta^2}{3} \end{array}\right\} \Rightarrow D\tilde{\theta}_1 = \frac{1}{n \cdot I(\theta)}$$

$\Rightarrow$ $\tilde{\theta}_1$ является эффективной оценкой

② $\tilde{\theta}_3'$ регулярность оценки

а) регулярность модели +

б) несмещённость оценки +

в) $D\tilde{\theta}_3' = \frac{13}{25}\theta^2$ ограничена на $\forall$ компакте из $(0; +\infty)$

$\Rightarrow$ оценка $\tilde{\theta}_3'$ является регулярной

$$D\tilde{\theta}_3' > \frac{1}{n \cdot I(\theta)}$$

$$\frac{13}{25}\theta^2 > \frac{\theta^2}{3}$$

ничего сказать нельзя по достаточному условию